RaPDTool v2.3.0

Rapid Profiling and Deconvolution Tool

Reference Manual

1. Overview

RaPDTool offers a simple, easy-to-use workflow for microbial-community profiling, contig binning, and "genomic-distance" exploration by chaining several bioinformatic tools into a single pipeline. It features robust error handling, parallelism, and automatic caching of external databases.

2. Run Modes (Architecture)

Global Input: Metagenomes or single genomes in FASTA format (.fasta, .fa, .fna, .fas, optionally .gz). The SCREEN mode also accepts raw FASTQ reads (.fastq, .fq).

FULL (Default)

Metagenome → MAGs

- **FOCUS** — Assembly taxonomic profile
- **Metabat2** — Contig binning into MAGs
- **Binning_refiner** — Refine to non-redundant set
- **miComplete** — Completeness & redundancy
- **Mash** — Genomic distance vs DB (Top 10 hits)

PROFILE

Single genome / assembly

- **FOCUS** — Assembly taxonomic profile
- **Mash (Whole Assembly)** — Classify assembly vs Type-Material DB

SCREEN

"What's there" • No binning

- **FOCUS** — Taxonomic profile (Assembly or Reads)
- **Mash Screen** — Reference-genome containment (≥ 0.95 id)

3. Installation

RaPDTool installs via Conda and runs a prebuilt, tested Apptainer image—no bioinformatics tools need to be installed on your host machine. The image and reference databases are downloaded and cached automatically on the first run.

```
conda create -n rapdtool -c conda-forge -c kjestradag rapdtool
conda activate rapdtool
rapdtool setup       # Optional: pre-fetch image + databases
rapdtool --where     # Show where image and databases are cached (~/.cache/rapdtool)
```

** External Databases: You can override the auto-downloaded databases by exporting* `$RTMASHDB` *(e.g., GTDB r202) and* `$RTFOCUSDB`.

4. Usage & Parameters

```
rapdtool -i INPUT [-o OUTPUT] [-m {full,profile,screen}] [-t THREADS] [-a COVERAGE]
         [--screen-identity F] [--no-split-bins] [--force] [-c COMMENT]

  -i, --input        Input FASTA (.fasta/.fa/.fna/.fas, opt. .gz). SCREEN accepts FASTQ. [Required]
```

```
  -o, --output       Output directory (default: ./rapdtool_results)
  -m, --mode         full (default): binning + per-bin taxonomic classification (MAGs);
                     profile: single-genome assembly (FOCUS + whole-assembly Mash);
                     screen: FOCUS + mash-screen (what reference genomes are present), no binning.
  -t, --threads      Threads for FOCUS/Metabat/miComplete/Mash (default: all cores)
  -a, --coverage     Depth/coverage file passed to Metabat2 (-a)
      --screen-identity  Min mash-screen identity in screen mode (default: 0.95)
      --no-split-bins    Disable per-species FASTA output
      --force        Overwrite existing results for the same input
  -c, --comment      Comment recorded in the log

  -d, --database     Mash .msh to use instead of the cached one (optional override)
      --focus-db     FOCUS db directory (containing db/k6) to use (optional override)
```

Examples

```
# Full pipeline (metagenome assembly)
rapdtool -i assembly.fasta -o results
# Screen: what reference genomes are present from raw FASTQs (no binning)
rapdtool -i reads.fastq.gz -m screen -o screen_out
# Profile a single-genome assembly
rapdtool -i genome.fna -m profile -o prof_out
# Full pipeline with 16 threads and a precomputed coverage file
rapdtool -i assembly.fa -t 16 -a depth.txt
```

5. Output Files & Examples

Results are written under the `-o` directory (default: `rapdtool_results`):

- F P S **rapdtool_confidence.tbl** / **.txt**: Merged high-confidence Genus & Species report. (Full mode adds completeness, redundancy & bin columns).
- F P S **rapdtool_krona.html**: Interactive Krona chart of the FOCUS taxonomic profile.
- F P S **profilesfmbm**/ and **allresultsfmbm**/: FOCUS profiling tables and the 10 closest Mash hits per bin. Includes execution logs.
- F **species_bins**/: One consolidated FASTA per identified species (e.g., <Species>__<bin>.fna), ready for downstream ANI/OGRI.
- F **workfmbm**/: Intermediate binning and distance data.

Example Output: miComplete (Bin Quality Check)

Found in `workfmbm/outmicomplete/` (Full mode). Evaluates the refined bins.

Bin_Name	Completeness	Redundancy	Marker_lineage
meta-assembly_bin_1.fna	98.50	1.25	Bacteria
meta-assembly_bin_2.fna	85.40	3.10	Archaea

Example Output: Mash (Taxonomic Neighborhood)

Found in `allresultsfmbm/`. For each bin, RaPDTool reports the ten closest neighbors. **Note:** A Genomic Distance < 0.05 strongly suggests genomic coherence (same species).

Query_genome	Match_in_database	Genomic_Distance	p_value	Shared_Hashes
meta-assembly_bin_1.fna	GCF_002165255.2 (Acinetobacter sp. WCHA45)	0.0620120	0	345/1000
meta-assembly_bin_2.fna	GCA_000430225.1 (Acinetobacter junii)	0.0327655	0	471/1000

6. Image Dependencies

FOCUS • Metabat2 (2.15) • Binning_refiner (1.4.3) • miComplete (1.1.1) • Mash (2.3) • KronaTools • entrez-direct.